FACULTY OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

JU HALL EVENT SYSTEM

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A RESEARCH PROJECT SUBMITTED IN PARTIAL FULFILMENTOF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF BACHELOR OF COMPUTER SCIENCE AND IT OF JAZEERA UNIVERSITY

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DECLARATION

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Title of Project Paper: JU Hall Event System

Field of Study: Mobile Application Development

We, the undersigned, do solemnly and sincerely declare that:

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Subscribed and solemnly declared before

Date: _________________________________

Supervisor's Signature: _____________________ **Date:** ___________________

DEDICATION

We dedicate this project to our beloved parents, brothers, sisters, and our entire families for their endless love, prayers, and encouragement throughout our academic journey.

We also dedicate this work to all our friends and colleagues who have supported us in one way or another.

Finally, we sincerely dedicate this research to our supervisor, **Eng. Abdirizak Mohamed Ahmed**, for his continuous guidance, patience, and valuable advice that made this project possible.

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Finally, we would like to sincerely thank our classmates, colleagues, and graduate students for their cooperation and encouragement. We also wish to express our gratitude to ourselves as a team, for supporting one another and working together to successfully complete this thesis.

APPROVAL SHEET

This is to confirm that the project entitled **"JU Hall Event System"** submitted by the under-mentioned candidates for partial fulfillment of the requirements of the **Bachelor Degree in Computer Science and IT**, under the Faculty of Computer Science and IT, Jazeera University, is an original research project. The results embodied in this project have not been submitted elsewhere for the award of any degree in this University or any other institution.

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LIST OF TABLES

LIST OF APPREVIATIONS

JU -Jazeera Univeversity

EMS - Event Management System

UI -User Interface

UX - User Experience

DB - Database

DFD - Data Flow Diagram

ERD - Entity Relationship Diagram

UML - Unified Modeling Language

UAT - User Acceptance Testing

RBAC - Role-Based Access Control

API - Application Programming Interface

SDK - Software Development Kit

CLI - Command Line Interface

NoSQL - Non-Structured Query Language

CRUD - Create, Read, Update, Delete

JSON - JavaScript Object Notation

IDE - Integrated Development Environment

OS - Operating System

BaaS - Backend as a Service

ICT - Information and Communication Technology

AI - Artificial Intelligence

UI/UX - User Interface / User Experience

CPU - Central Processing Unit

RAM - Random Access Memory

SSD - Solid-State Drive

PDF - Portable Document Format

IT - Information Technology

SMS - Short Message Service

TABLE OF CONTENTS

CHAPTER ONE

INTRODUCTION

1.0 Introduction

In the digital age, the integration of technology into university operations has become essential for enhancing efficiency, transparency, and service delivery. Among the many areas benefiting from digital transformation is the management of university facilities—particularly halls and event spaces, which are in constant demand for academic, administrative, and social activities (Laudon & Laudon, 2022).

At Jazeera University, hall reservations have traditionally been managed through manual processes, involving paperwork, physical visits, and time-consuming approval chains. This outdated approach has led to challenges such as double bookings, scheduling conflicts, lack of real-time availability tracking, and limited access for students and staff (Unwin, 2019).

To address these issues, this project introduces a modern, mobile-based event hall system designed specifically for Jazeera University. The proposed application enables users to request and reserve halls through an intuitive mobile interface, while administrators can monitor bookings, approve or reject requests, and oversee hall usage in real-time. Features such as role-based access, calendar integration, push notifications, and cloud synchronization are incorporated to ensure a seamless and reliable experience for all stakeholders (Zhou, Zhang & Zimmermann, 2018).

Globally, event and hall booking systems have progressed from manual, paper-based methods in the 1990s to web-based platforms in the 2000s, and today to mobile and cloud-based solutions that provide real-time access and automation (Marinescu, 2017).

Regionally, East African universities such as Nairobi and Makerere have adopted centralized digital booking systems to reduce scheduling conflicts and improve efficiency, though challenges remain with infrastructure and cost (Gupta & Kohli, 2006).

Locally, most Somali universities, including Jazeera, still depend on manual processes, making the introduction of a mobile-based hall booking system both timely and pioneering (Ally & Tsinakos, 2014).

1.1 Background of the Problem

Universities are vibrant institutions that continuously organize a wide range of events such as academic lectures, conferences, examinations, workshops, seminars, cultural festivals, and administrative meetings. Managing these events effectively is crucial to the smooth functioning of higher education institutions. Traditionally, universities have relied on manual approaches to schedule and allocate halls for events. These include the use of paperwork, noticeboards, and verbal communication, which are often time-consuming and susceptible to errors (Laudon & Laudon, 2022). As a result, issues such as double booking of lecture halls, scheduling conflicts, and inefficient utilization of resources are common problems that negatively affect the academic environment.

The adoption of Event Management Systems (EMS) in universities offers a solution to these challenges. EMS platforms automate hall booking, event scheduling, and resource management, ensuring that students and staff can easily access real-time information about room availability and event details. Research shows that universities implementing web-based or cloud-based event management solutions have reported improved efficiency, better coordination, and enhanced transparency in resource allocation (Marinescu, 2017; Gupta & Kohli, 2006).

In recent years, cloud computing has become an enabler of modern EMS by providing scalable infrastructure, centralized storage, and multi-user access. Cloud-driven event management allows stakeholders to update and retrieve event information simultaneously, minimizing the risk of miscommunication (Zhou et al., 2018). Furthermore, cloud solutions reduce the financial burden of hardware maintenance for universities, which is particularly significant in resource-constrained environments (Unwin, 2019).

Alongside cloud solutions, mobile-first event management systems have become popular in higher education. Mobile applications allow students and staff to check schedules, book halls, and receive notifications in real-time. Since mobile devices are widely used by students, mobile-first design ensures inclusivity and accessibility (Ally & Tsinakos, 2014). The use of frameworks such as Flutter for cross-platform development and Firebase for backend services further supports the development of cost-effective and efficient EMS (Kumar & Rathi, 2021).

The acceptance of such digital systems is supported by theories like the Technology Acceptance Model (TAM) and Unified Theory of Acceptance and Use of Technology (UTAUT), which emphasize that user perceptions of usefulness and ease of use are critical in determining the adoption of new technologies (Davis, 1989; Venkatesh et al., 2003). For universities in developing countries, where infrastructural and financial challenges are common, adopting EMS can significantly improve operational efficiency, transparency, and user satisfaction, making them essential tools for modern higher education.

1.2 Problem Statement

Jazeera University currently lacks an efficient, transparent, and user-friendly system for managing the scheduling and booking of university halls. The existing manual process, which relies on paper forms and in-person approvals, has become increasingly unsustainable as the number of academic and extracurricular events continues to grow. This outdated system causes frequent scheduling conflicts, miscommunication, and delays in event coordination.

Studies and internal observations have shown that hall bookings at the university are often duplicated, misplaced, or approved late. These issues have been consistently reported by both administrative staff and student organizations. Moreover, the demand for campus space has increased steadily in recent years, making timely and accurate hall scheduling more important than ever.

The lack of a centralized, real-time booking platform has exacerbated these challenges. Unlike many institutions that have embraced digital tools for event management, Jazeera University still depends on paper logs and basic spreadsheets— tools which do not scale well in a busy academic environment. This trend reflects a broader gap between the university's current operations and modern digital campus practices.

Key terms in this problem include:

1.Hall Booking System: Platform for requesting and managing event spaces.

2.Real-Time Availability: Immediate access to hall usage status.

3.Role-Based Access: Different levels of access for admins and users.

4.Double Booking: Two or more events scheduled for the same hall and time**.**

The root causes of the problem include:

1.Reliance on manual paperwork and human coordination

2.Absence of automated approval and notification system

3.Lack of data tracking for room utilization

4.No mobile access for students and staff

Therefore, there is a pressing need for a centralized, scalable, and real-time digital hall management system. Such a solution would not only streamline the booking process but also improve communication, reduce conflicts, and support datadriven decision-making within Jazeera University.

These limitations underline the need for a scalable and reliable digital solution that meets the needs of both students and administrators.

1.3 Purpose Of the Project

The purpose of this project is to develop a mobile application that automates the hall booking and event management process at Jazeera University. The system is designed to allow users including external guests or organizations—to request hall reservations, check availability, and receive booking confirmations in a seamless and user-friendly manner.

Administrators can manage hall schedules, approve or decline booking requests, and monitor hall usage through an integrated analytics dashboard.

The mobile application aims to enhance operational efficiency, reduce scheduling conflicts, and support the university's increasing demand for space by providing a modern digital platform that caters to both internal and external users.

1.4 Research Questions

This study seeks to answer the following research questions:

1.What are the key problems in the current manual hall booking process at Jazeera University?

2.What challenges do students and staff face when using the manual system?

3.How can a mobile-based system improve efficiency and user satisfaction?

4.What essential features should the proposed hall booking system include for Jazeera University?

1.5 Motivation of the Study

The motivation for this study arises from the researcher's observation of frequent scheduling conflicts, double bookings, and inefficient manual processes in managing university halls at Jazeera University. As the number of events and activities on campus continues to increase, the absence of a reliable digital solution has led to wasted time, miscommunication, and dissatisfaction among students and staff.

This project is driven by the desire to modernize Jazeera University's hall booking process by introducing a mobile-based system that aligns with global digital practices while addressing local challenges. The researcher is also motivated by the need to reduce administrative workload, improve transparency, and provide real-time access to booking services for both staff and students.

Furthermore, the study is inspired by the growing importance of digital transformation in higher education institutions, where technology is seen as a key enabler for efficiency, inclusiveness, and better service delivery. Implementing such a system at Jazeera University would not only enhance operations but also serve as a model for other Somali universities.

1.6 Research Objectives

The specific objectives of this project are:

1.To analyze and evaluate the existing manual hall booking process at Jazeera University for improvement

2.To design and develop a mobile-based hall booking and event system that is user-friendly and accessible to both internal users (students, staff) and external guests (organizations, alumni).

3.To implement role-based access control, ensuring that different types of users have appropriate levels of system access and functionality.

4.To enable real-time hall availability tracking and automated notifications using cloud-based technologies such as Firebase, improving responsiveness and reducing scheduling conflicts.

5.To provide an administrative dashboard with analytics tools that allow university management to monitor hall usage, track booking patterns, and generate insightful reports for decision-making.

6.To test and evaluate the system's performance, usability, and effectiveness through user feedback, ensuring it meets the operational needs of Jazeera University.

1.7 Significance of the Study

This project is significant in several ways: For university staff and administrators: It provides a fast and organized way to manage hall booking requests, improve space utilization, and streamline event approval processes through a centralized digital system. For external users: It allows guests, alumni, and external organizations to request hall reservations for events such as conferences, weddings, seminars, or community celebrations, expanding access beyond internal use and enhancing public engagement with the university. For event attendees such as students and visitors: It ensures better coordination, accurate scheduling, and timely communication regarding hall-based activities thus improving the overall event experience.

1.8 Scope & Limitations

This project focuses on designing and implementing a mobile-based hall booking system for Jazeera University. The system will allow registered users to book, view, and manage halls in real time, while administrators handle approvals and monitor scheduling. Role-based access will distinguish between regular users and administrators to ensure efficient and organized operations.

1.8.2 Limitations

Despite its broad usefulness, the project has the following limitations:

1. The system is limited to Jazeera University only.

2. It focuses solely on hall booking, not other university services.

3. Real-time access depends on stable internet connectivity.

4. Full effectiveness may be affected by user adoption and institutional policies.

1.9 Organization of the study

This chapter has provided a comprehensive introduction to the hall event management application at Jazeera University, including the background, problem statement, purpose, objectives, scope, and significance. It has demonstrated the need for a digital solution that addresses the limitations of the university’s current manual hall booking process.

1.10 Chapter Summary

This chapter presented the overall foundation of the study by introducing the concept, background, and necessity of developing a mobile-based Hall and Event Management System for Jazeera University. It discussed how the increasing demand for efficient hall reservation processes has made manual systems outdated, leading to challenges such as double bookings, poor coordination, and delays in approval.

The background section explained how digital transformation—through technologies such as cloud computing, mobile applications, and real-time data systems—can improve efficiency, transparency, and accessibility in university operations. The problem statement identified the limitations of the current manual booking system, while the purpose and objectives outlined the project's aim to design a scalable, user-friendly, and real-time mobile application for managing university halls and events.

Additionally, the chapter highlighted the research questions, motivation, significance, scope, and limitations of the study, showing how the proposed system will benefit students, administrators, and external users by improving communication, space utilization, and service delivery.

Chapter Two offers an in-depth literature review related to event management systems, presents theoretical and conceptual frameworks, compares existing technologies, and identifies gaps in current research.

Chapter Three details the software development methodology adopted in building the system, including requirement gathering, analysis tools, modeling techniques such as DFD and UML, and feasibility studies.

Chapter Four focuses on the design aspects of the system, including architecture, user interface, database schema, data dictionary, and forms and reports necessary for operation.

Chapter Five explains the implementation process of the developed system, describes the testing procedures conducted (unit, integration, and acceptance), and presents results from user feedback and system evaluation.

Chapter Six concludes the project by summarizing the major findings, highlighting the system's strengths and limitations, and offering recommendations for future improvements and scalability.

CHAPTER TWO

LITERATURE REVIEW

2.0 Introduction

The chapter reviews scholarly literature, theories, and technologies related to event and hall systems in higher education. The review aims to build an academic foundation for the development of a mobile-first, cloud-based event management system tailored for Jazeera University.

According to Creswell (2018), a literature review is not merely a summary of past studies but also a critical synthesis that identifies research gaps. For this reason, the review begins with theoretical frameworks such as the Technology Acceptance Model (TAM) and Diffusion of Innovation (DOI), followed by a discussion of related systems, technological advancements (cloud computing, Flutter, Firebase), and their relevance to university event management.

2.1 Theoretical Framework

The development of the Jazeera Hall Event App is underpinned by established theories in Information Systems and technology adoption. These frameworks provide both academic justification and practical guidance for system design.

a) Information Systems Theory

Information Systems (IS) Theory explains how data flows through organizations and how technology can enhance decision-making, efficiency, and productivity (Laudon & Laudon, 2022). For Jazeera University, the hall booking process involves multiple stakeholders administrators staff and students who require accurate and timely information.

An IS-based system improves coordination, reduces conflicts, and supports informed decisions.

b) Technology Acceptance Model (TAM)

Davis (1989) introduced the TAM, which suggests that perceived usefulness and ease of use are critical factors influencing the adoption of technology. Applying TAM to this study emphasizes the importance of developing a user-friendly mobile interface that students and staff can easily adopt without technical barriers.

c) Diffusion of Innovation (DOI) Theory

Rogers (2003) highlights how innovations spread within organizations and societies. The adoption of a mobile-first hall booking system represents an innovation in the Somali higher education context. Factors such as relative advantage, compatibility, and observability will influence its acceptance.

d) Cloud Computing and Mobile-First Models

Modern system design emphasizes cloud-based infrastructure and mobile-first principles. Zhao et al. (2020) argue that real-time synchronization and scalability are vital in academic systems. Mobile-first models ensure accessibility for students and staff who primarily use smartphones to access university services.

Framework Implications[Text Wrapping Break]Together, these theories provide a conceptual model for the Jazeera Hall Event App. The model emphasizes:

User Accessibility: Mobile-first interfaces for students and staff.

Real-Time Synchronization: Always updated hall availability and booking status.

Role-Based Access: Differentiated permissions for administrators, organizers, and general users (Kim et al., 2020).

2.2 Related Work

Several systems have been developed to streamline event and hall booking processes in universities. This section reviews related works, identifying strengths, limitations, and lessons for the proposed system.

Review of Related Systems

System Name	**Technology Used**	**Key Features**	**Limitation**	**Gaps addressed by proposal system**
Kim et al. (2020) – University Room Reservation	Web-based, MySQL Database	Web interface Basic booking Conflict reduction	No mobile access No real-time updates	Mobile-first design - Realtime synchronization

Ahmed & Salim (2022) – Smart Event Booking App	React Native, Firebase	Mobile scheduling- Instant notifications	No admin panel Single-role users only	Role-based access Admin controls
Proposed Jazeera Hall Event App	Flutter, Firebase	Mobile-first interface	N/A (designed to overcome previous limit	All above gaps resolved

Table 2.1 Review of related systems

Commercial Comparisons

many universities develop in-house systems, commercial tools like Eventbrite, Google Calendar, and Microsoft Teams Booking offer scheduling solutions. However, these systems lack customization for academic contexts and do not provide role-based academic administration features.

Use of Flutter and Firebase in Mobile App Development

Flutter, developed by Google, is a robust cross-platform framework that enables developers to build responsive applications for both Android and iOS using a single codebase (Kumar & Rathi, 2021). Its widget-based architecture ensures fast UI rendering and a native-like experience.

Firebase, on the other hand, provides backend-as-a-service (BaaS) solutions, including authentication, cloud storage, Firestore real-time databases, and push notifications. Integration with Flutter ensures seamless synchronization of hall booking data, which is critical for resolving conflicts and improving efficiency (Zhao et al., 2020).

2.3 Gaps in Existing Literature

Despite the presence of numerous event management systems, several critical gaps remain—particularly in the higher education context of developing countries such as Somalia:

1.Over-Reliance on Manual Systems[Text Wrapping Break]Many universities continue to use paper-based processes or noticeboard announcements for hall reservations. This reliance on manual systems creates inefficiencies, scheduling errors, and communication breakdowns, limiting overall productivity (Laudon & Laudon, 2022).

2.Lack of Real-Time Booking Several existing systems provide basic digital booking but do not offer real-time synchronization. This absence of live updates often results in double-booking of lecture halls and conflicts between administrators and students (Zhou, Zhang, & Zimmermann, 2018).

3.Absence of Role-Based Access Studies indicate that most web-based reservation frameworks do not differentiate between administrators, organizers, and students. Without role-based permissions, administrative control is weakened and security risks increase (Kim, Park, & Lee, 2020).

4.Delayed Approvals and Communication Many event management platforms lack automated notification systems and approval workflows. As a result, users face delays in receiving confirmation, which reduces satisfaction and efficiency (Gupta & Kohli, 2006).

5.Limited Mobile Accessibility Although smartphones are the primary means of internet access for students in many universities, numerous existing systems are designed with a “web-first” approach, which reduces accessibility for mobile users (Ally & Tsinakos, 2014).

6.Lack of Local Customization Commercial event management platforms such as Eventbrite or Google Calendar are designed for corporate or Western educational contexts. These tools are not adapted to the specific needs of universities in developing regions, including Somali institutions (Unwin, 2019).

7.No Data Analytics Current systems rarely provide administrators with dashboards or reporting tools to analyze hall usage and track trends. The absence of analytics limits evidence-based decision-making in resource management (Gupta & Kohli, 2006).

2.4 Critical Review & Gap Identification

The critical review of past systems reveals that while significant progress has been made in automating hall and event booking, important shortcomings persist in higher education event management.

Dependence on Manual Systems Many universities in developing countries still rely heavily on manual hall booking methods such as paperwork and noticeboards. This dependence results in inefficiencies, scheduling errors, and miscommunication (Laudon & Laudon, 2022).

Lack of Real-Time Updates Existing digital systems often fail to provide real-time synchronization of booking data, which leads to double bookings and conflicts in scheduling (Zhou, Zhang, & Zimmermann, 2018).

Insufficient Role-Based Access Several web-based reservation frameworks do not include differentiated access for administrators, lecturers, and students. This undermines administrative control and creates security concerns (Kim, Park, & Lee, 2020).

Delayed Approvals and Communication In many systems, communication between users and administrators is not automated. This lack of instant notifications and approval workflows delays the process and reduces user satisfaction (Zhou et al., 2018).

Limited Mobile Accessibility Although mobile devices are widely used among students, many existing solutions are web-first designs. This restricts accessibility and makes adoption challenging in student communities (Ally & Tsinakos, 2014).

Inadequate Customization for Local Context Most commercial systems (e.g., Eventbrite, Google Calendar) are designed for corporate or Western academic environments. They lack localization features required in Somali higher education, making them less effective (Unwin, 2019).

Absence of Data Analytics Current event scheduling systems rarely provide administrators with dashboards or analytical tools to evaluate hall utilization, track trends, or generate performance reports. This absence prevents data-driven decision-making (Gupta & Kohli, 2006).

2.5 Research Gap

These gaps indicate a strong need for a mobile-first, cloud-based, and role-sensitive system tailored specifically for Jazeera University. Unlike general-purpose solutions, the proposed Jazeera Hall Event App is designed to address the shortcomings of existing systems by providing:

Real-Time Hall Booking Updates: To eliminate double-booking and scheduling conflicts, ensuring reliable information sharing across all stakeholders (Zhou, Zhang, & Zimmermann, 2018).

Differentiated Access for Administrators, Organizers, and Students: A role-based access control system improves security, accountability, and administrative efficiency (Kim, Park, & Lee, 2020).

Push Notifications and Instant Approvals: Automated alerts and approval workflows streamline communication and improve user satisfaction by reducing delays (Gupta & Kohli, 2006).

Customization for the Somali Academic Context: Unlike commercial solutions designed for Western institutions, a locally customized system can better meet the unique cultural, linguistic, and institutional requirements of Somali universities (Unwin, 2019).

Data Analytics Dashboards for Management: Providing decision-makers with analytical tools supports evidence-based planning, performance monitoring, and efficient resource utilization (Laudon & Laudon, 2022).

2.6 Chapter Summary

This chapter has reviewed the theoretical foundations, related works, and existing technologies relevant to hall booking systems in universities. It highlighted how Information Systems Theory, TAM, and Diffusion of Innovation support the adoption of mobile-first solutions. The review of related systems demonstrated that while previous works have introduced digital solutions, they remain limited in terms of mobile accessibility, real-time synchronization, role-based access, and local customization.

The chapter also identified research gaps that justify the development of the Jazeera Hall Event App. By addressing these limitations, the proposed system will enhance efficiency, improve communication, and align with the specific needs of Jazeera University.

The next chapter (Chapter Three) will focus on the methodology, including system design, user requirements, feasibility study, and modeling tools such as Data Flow Diagrams (DFD) and Unified Modeling Language (UML).

CHAPTER THREE

METHADOLOGY

3.0 Introduction

The chapter explains how the Jazeera University Hall Event mobile application was developed. It describes the steps and methods used to build the system, starting from requirement gathering to design, development, and testing.

The development followed a clear methodology to ensure that the final product is useful, reliable, and meets the needs of its users. The app was built using **Flutter**, which allows it to work on both Android and iOS devices with a single codebase. **Firebase** was used for the backend, providing real-time data storage, user authentication, and easy system scalability.

3.1 Research Design

The study adopts a **design and development research approach**, focusing on creating a functional mobile application tailored to Jazeera University's hall booking and event management needs. This approach combines theoretical research with practical implementation, ensuring both academic rigor and real-world applicability.

3.2 System Description

The system is a mobile-first hall booking application that allows students, staff, and external users to book halls, receive notifications, and view hall availability. Administrators manage approvals, schedules, and generate usage reports.

3.3 Overview of the System

The app consists of the following main modules:

1.**User Module:** Booking requests, notifications, view hall availability.

2.**Admin Module:** Approve/reject bookings, manage schedules, access analytics.

3.**Database Module:** Stores users, events, halls, and bookings.

3.4 System Features

1.Real-time booking and updates.

2.Role-based access control.

3.Mobile-first interface with cloud integration.

4.Notifications for approvals and updates.

5.Analytics for hall usage.

3.5 System Development Environment

1.Programming Tool: Flutter, Dart, Firebase CLI

2.Backend & Database: Firebase Authentication, Firestore

3.Operating System: Windows 10/11

4.Browsers: Chrome, Edge, Safari, Firefox

5.Hardware: Development computers (min 8GB RAM, 256GB SSD), mobile devices for testing

Operational Framework

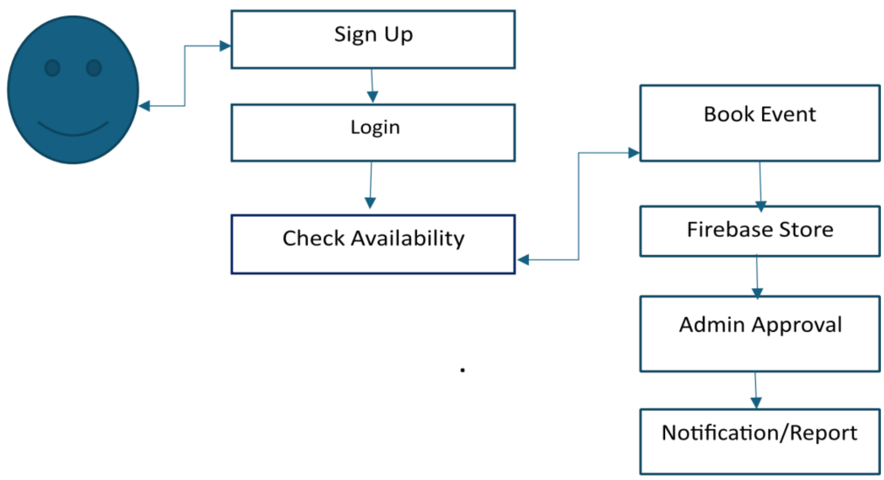

Figure3.1 Operational framework

3.6 System Requirement

3.6 System Requirement Specification

3.6.1 Software Requirements

No	REQUIREMENT	Description
1	Operating System	Windows 10/11
2	Browser	Chrome, Edge, Safari, Firefox
3	Programming Tool	Flutter, Dart, Firebase CLI
4	Backend & Database	Firebase,Firestore, Firebase Auth

Table3.1 Software requirements

3.6.2 Hardware Requirements

No	Item Name	Quantity	Description
1	Development Computers	4x	Laptops/desktops for developers Min: 8GB RAM, 256GB SSD OS: Windows/macOS/Linux
2	Mobile Devices	2x	Android or iOS smartphones Used for testing the Flutter app

Table3. 2Hardware requirements

3.8 User Requirements

The users of the Jazeera University Hall Event App are expected to meet the following requirements:

a) **Device Access:** Users should have access to a device such as a smartphone, tablet, or computer with a reliable internet connection.

b) **Basic Application Knowledge:** Users must possess basic knowledge of using mobile or web applications, including the ability to log in, book events, and receive notifications.

c) **Navigation and Comprehension:** Users should be able to read and follow on-screen instructions provided through the app interface to ensure successful navigation and usage.

3.9 Problems in the existing system include:

Manual booking process

Lack of centralized calendar

Double-bookings

No mobile interface for accessibility

The solution aims to address these challenges through a responsive, cloud connected mobile app.

3.10 Requirements Gathering Techniques

The following techniques were used:

Interviews with students and university staff

Questionnaires distributed to event organizers

Observational studies of current booking processes

Document analysis of hall usage logs

3.11 Use case Analysis

Use case analysis helps to identify and define the key functions that the proposed hall event system should perform. It describes how different users (actors) interact with the system to accomplish specific goals such as booking a hall, approving a request, or viewing availability.

The main actors in this system are the **User**, **Admin, each** actor interacts with the system differently based on their role and privileges. For example, users can submit booking requests and receive notifications, while administrators can manage hall schedules, approve or reject bookings, and access usage reports.

The following diagram illustrates the primary use cases and how each actor engages with the system.

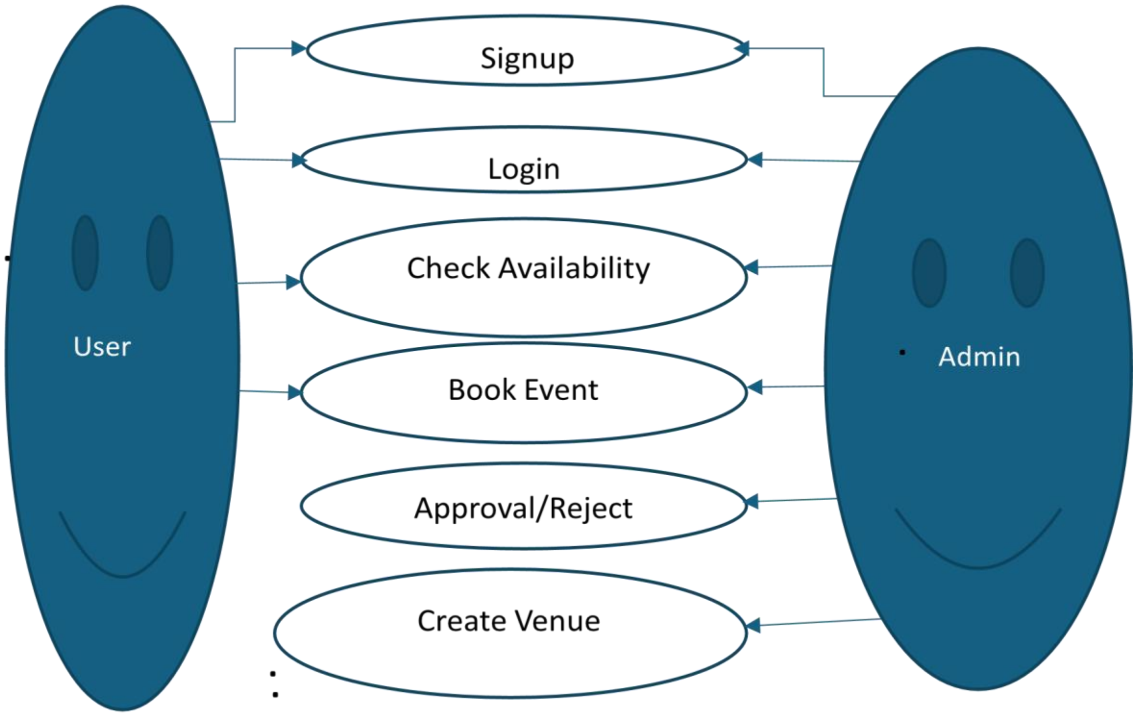

Figure 3.2 Use case diagram

Figure3. 2 Use case diagram

3.12 Data Flow Diagrams

Level 0 DFD (Context Diagram)

The Level 0 DFD (Context Diagram) represents the overall system as a single process called "Hall Booking System." It shows the interaction between external actors (Userss and Administrators) and the system, including the data they send and receive.

All major inputs and outputs are captured at this top level without internal details.

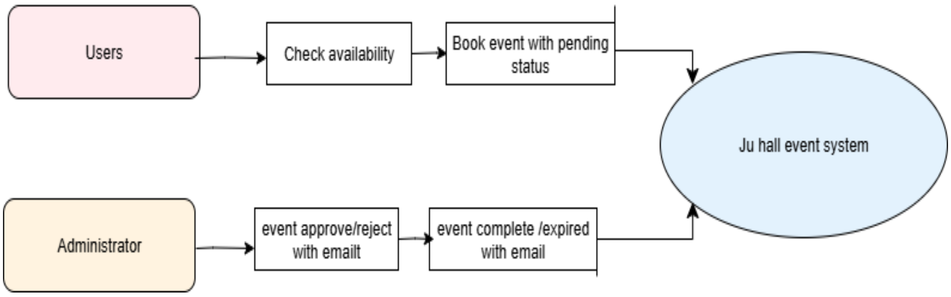

Figure 3.3DFD Level0 diagram

DFD Level 1

The Level 1 Data Flow Diagram illustrates the major functional processes of the Hall Event System used at Jazeera University. This diagram expands upon the context diagram by breaking down the system into four main processes and showing how data flows between them, the external entities (users), and the system's data stores.

The external entities interacting with the system are the User and the admin. The diagram captures how these users interact with the system and how data is exchanged with the system's internal components.

Process Breakdown:

Login & Authentication (Process 1.0)

Students or Admins initiate the process by submitting their credentials. The system then validates these credentials by checking the Users Database (Users DB). If the information is valid, access is granted.

Submit Booking Request (Process 2.0)

After login, a student can submit a booking request by filling out the booking form. The booking data is saved in the Bookings Database (Bookings DB), and a notification is sent to the admin to review the request.

Review & Approve/Reject Booking (Process 3.0)

The Admin reviews incoming booking requests, then either approves or rejects them. This decision is updated in the Bookings DB, and a corresponding notification is sent back to the student informing them of the outcome.

View Hall Availability (Process 4.0)

Students can view the availability of university halls by sending a request to the system. The system checks existing booking records from the Bookings DB and returns a list of available halls to the student.

This Level 1 DFD provides a clear, logical structure of how different components of the system communicate and how data flows through various user interactions.

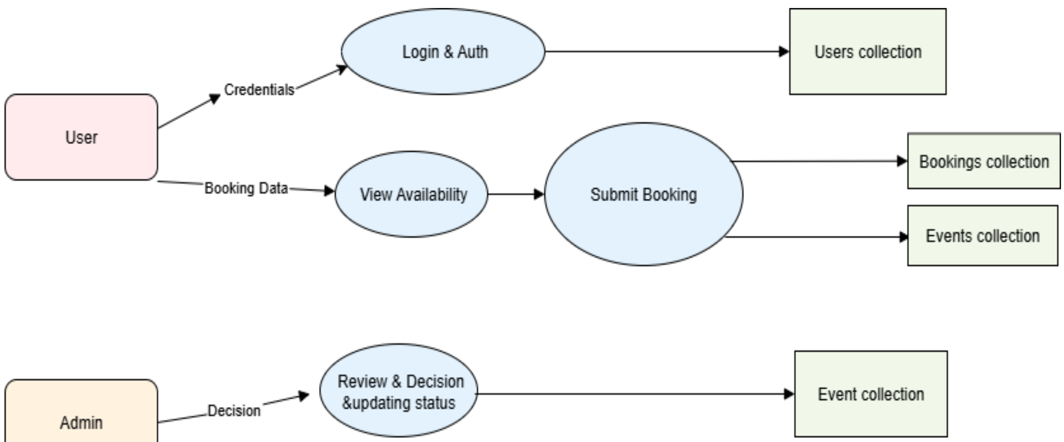

Figure3. 4 DFD Level1 diagram

3.13 ERD diagram

The Entity Relationship Diagram (ERD) illustrates the core structure of the hall booking system. It shows the relationships between four main entities: User, Hall, Event, and Booking.

A User can create events and make bookings.

.An Event is organized by a user and is held at a specific hall.

A Booking connects a user, an event, and a hall, indicating who booked which hall for which event.

A Hall can host multiple events and bookings.

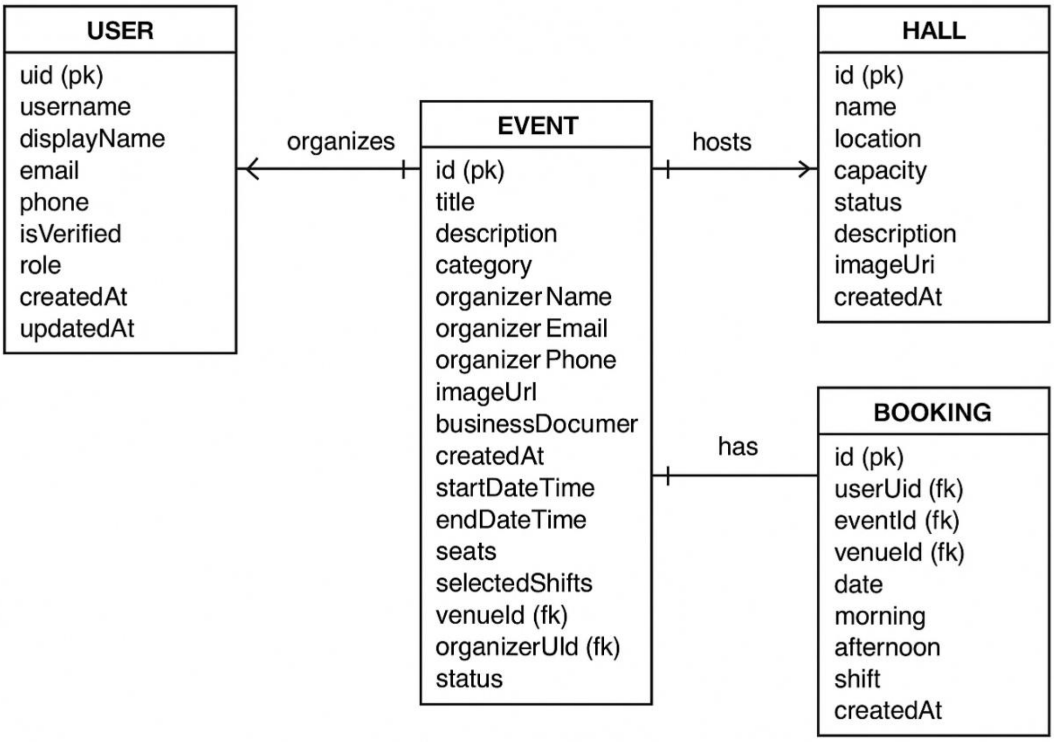

Figure3. 5 ERD diagram

3.14 Timeline of the project

No	Type	Time
1	Planning And Analyzing phase	2 months
2	Design Phase	2 months
3	Coding Phase	4 months
4	Testing Phase	2 months
Total	Total	12 months

Table 2.3Timeline of the project

This timeline was achieved with minor adjustments and no critical delays, confirming the system’s schedule feasibility.

3.15 Chapter Summary

This chapter detailed the software development methodology for the Jazeera University Hall Event app. It covered the requirement gathering process, system analysis, modeling, and feasibility considerations leading to the development of a mobile application tailored to the university's needs.

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CHAPTER FOUR

ANALYSIS AND DESIGN

4.0 Introduction

The chapter describes the software design of the Jazeera University Hall Event App. It focuses on how the application was architected, how the user interface was constructed, and how the system's data is modeled and managed using Firebase. This phase is essential for ensuring that the developed system meets user needs in terms of performance, usability, and maintainability.

This section analyzes the current hall reservation challenges at Jazeera University and highlights why a mobile-first solution is required. Manual processes and lack of real-time communication often lead to scheduling conflicts, delays, and poor user experience.

4.1 Architectural Design

The system follows a layered architecture where different components interact in a clear and structured manner.

1.At the top level, Mobile Users (both regular users and administrators) interact with the application through a user-friendly interface.

2.The Flutter Frontend acts as the bridge between users and the backend services. It handles both the user interface (UI) and the core logic of the application, such as user input, navigation, and state management.

3.The frontend communicates with three main backend services provided by Firebase:

4.Firebase Authentication: Manages user login, registration, and role-based access (user vs admin).

5.Firebase Firestore: A cloud-based NoSQL database that stores structured data such as event details, venue listings, and booking records.

6.Firebase Storage: Used to store media assets like event and venue images, allowing easy retrieval and display within the app.

This architecture ensures a smooth flow of data, scalability, and secure user management, making it ideal for mobile event management applications like the Jazeera University Hall Event App.

4.2 Existing Approaches

Several institutions use paper-based systems or basic web forms. While these provide partial solutions, they lack mobile accessibility, real-time updates, and role-based access. Such gaps confirm the need for a customized mobile application supported by Firebase cloud infrastructure.

4.3 The Proposed System

The proposed Jazeera University Hall Event App provides:

1.Mobile-first booking.

2.Real-time synchronization using Firebase Firestore.

3.Role-based access (Users vs. Admins).

4.Notifications for approvals and updates.

5.Analytical reports for decision-making.

4.4 Requirements

4.4.1 Functional Requirements

1.User registration and authentication.

2.Hall booking and availability checking.

3.Admin approval/rejection of bookings.

4.Notifications for status updates.

5.Event and venue management.

4.4.2 Non-Functional Requirements

Performance: Fast response time under heavy usage.

Scalability: Support for future expansion.

Security: Role-based access and encrypted communication.

Usability: Intuitive and consistent UI.

Reliability: Real-time synchronization and data backup.

4.5 System Design

Architectural Design

The system follows a layered architecture:

Mobile Users (Users/Admins): Interact via a Flutter-based UI.

Web Users (Users/Admins): Interact via a Flutter-based UI.

Flutter Frontend: Manages navigation, booking logic, and state.

Firebase Services:

1.Authentication for login and role control.

2.Firestore for event, venue, and booking data.

3.Storage for images.

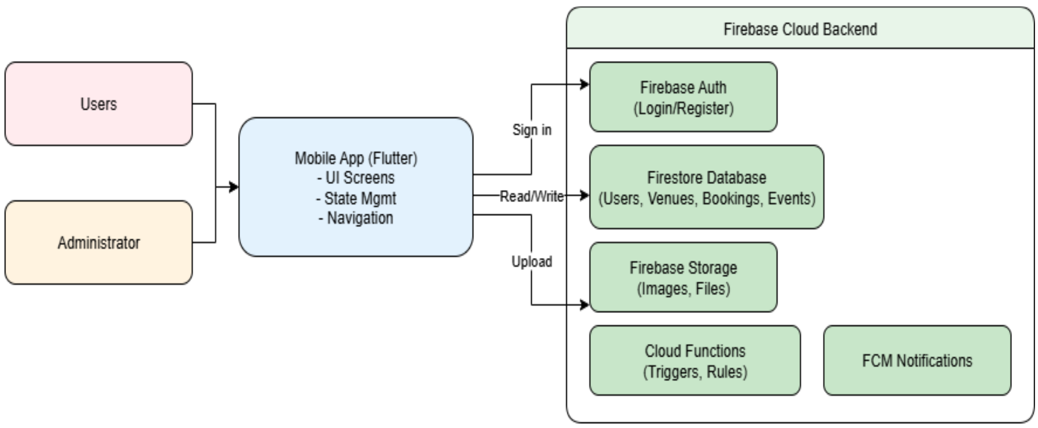


Figure4.1 Architectural diagram

4.6 USER INTERFACE DESIGN

The user interface (UI) design of the "Jazeera University Hall Event App" was carefully crafted to provide a smooth, intuitive, and accessible experience for both regular users and administrators. The UI was built using Flutter, allowing for a responsive and cross-platform layout that adapts well to various screen sizes.

The design of the interface followed key UI principles:

1.Simplicity – Minimalistic and clutter-free screens to reduce user confusion.

2.Consistency – Uniform design elements across all pages for better familiarity.

3.Accessibility – Large buttons and readable text for all users.

4.Responsiveness – Layouts adapt to different devices (phones, tablets).

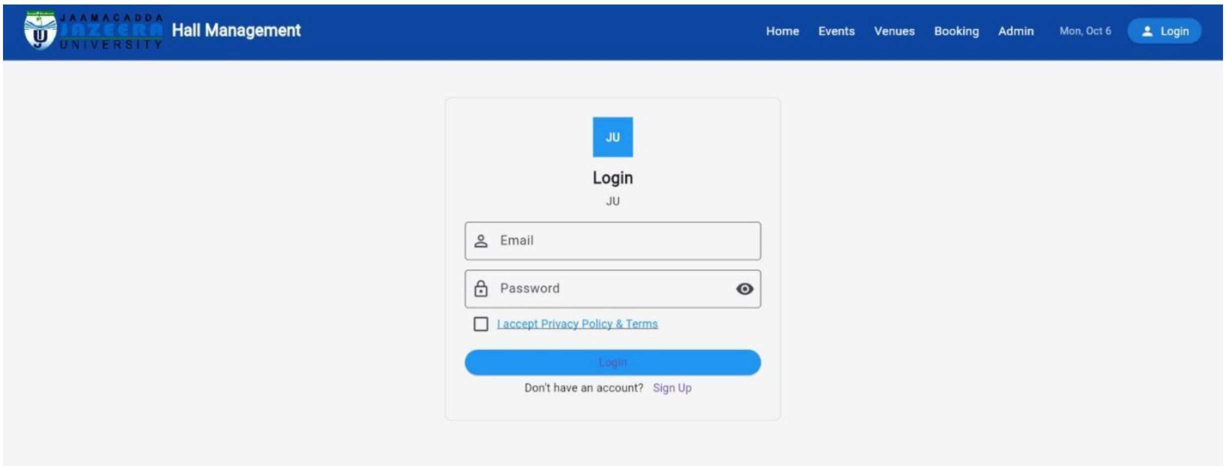


Figure4.2*Login Screen*

This screen shows the **Login Interface**, which provides:

Fields for email and password entry.

A checkbox to accept Privacy Policy & Terms.

An option for new users to create an account (Sign Up).

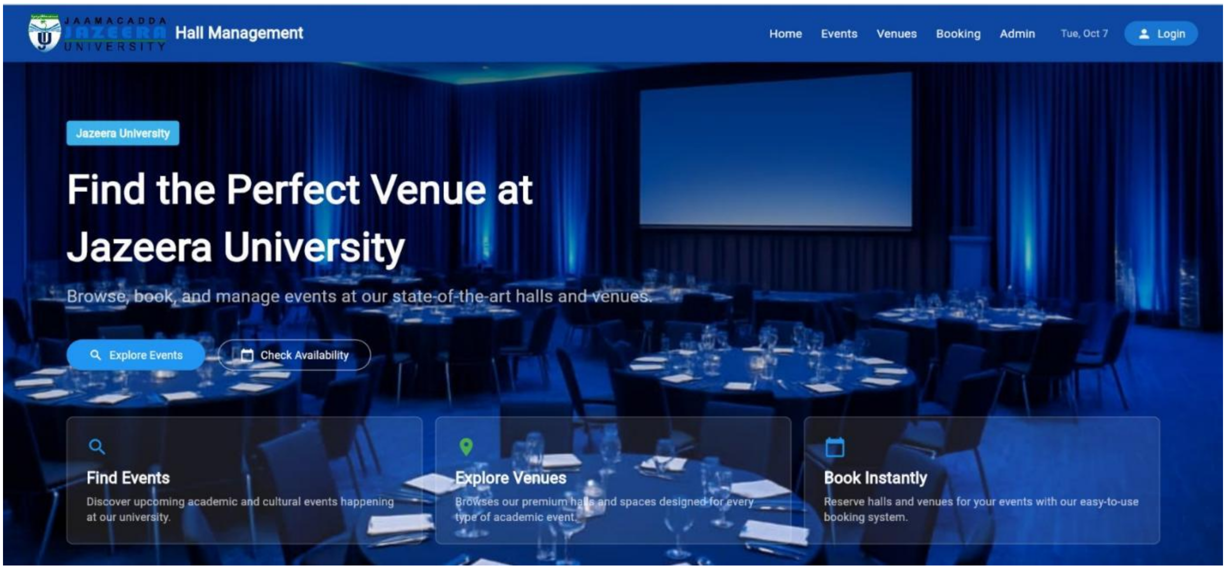


Figure 4.3Home interface

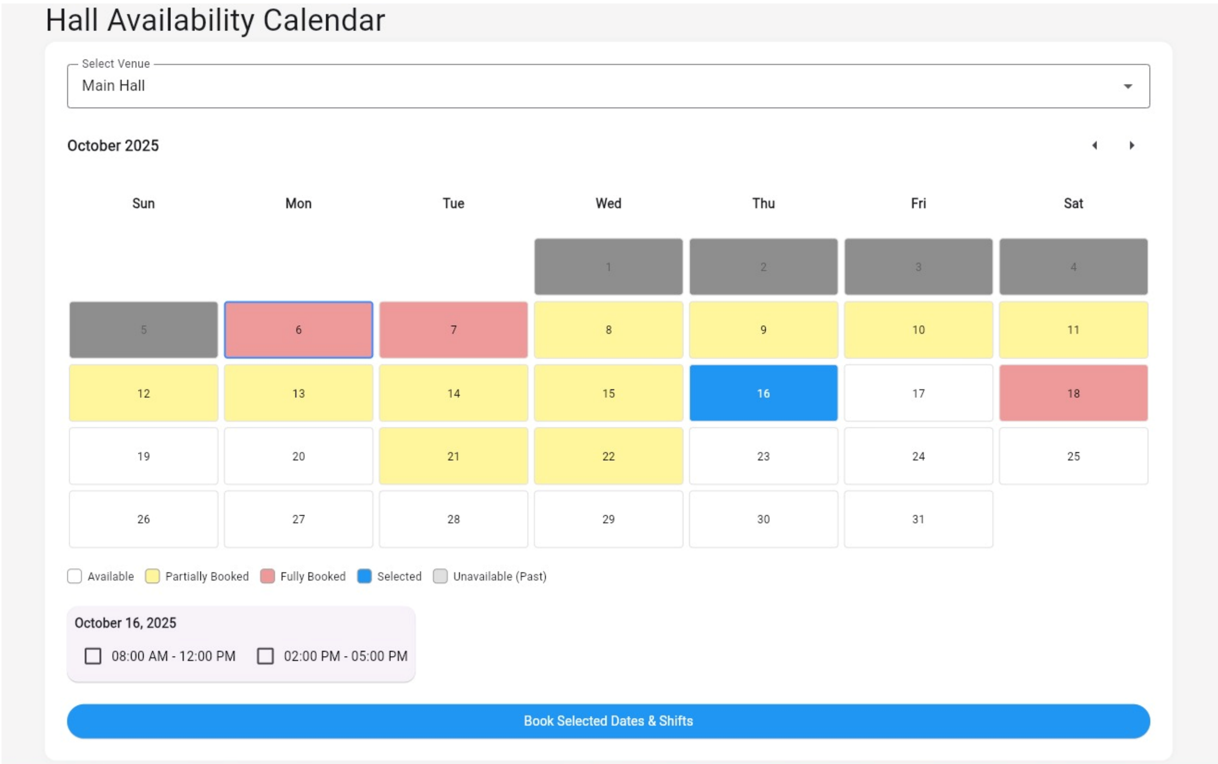


Figure 4.4Booking Interface

The Availability Calendar allows users to check and book halls by date. It features:

Color codes indicating availability (Available, Partially Booked, Fully Booked, Selected, Past/Unavailable).

The ability to select time slots for bookings.

A simple and interactive booking process.

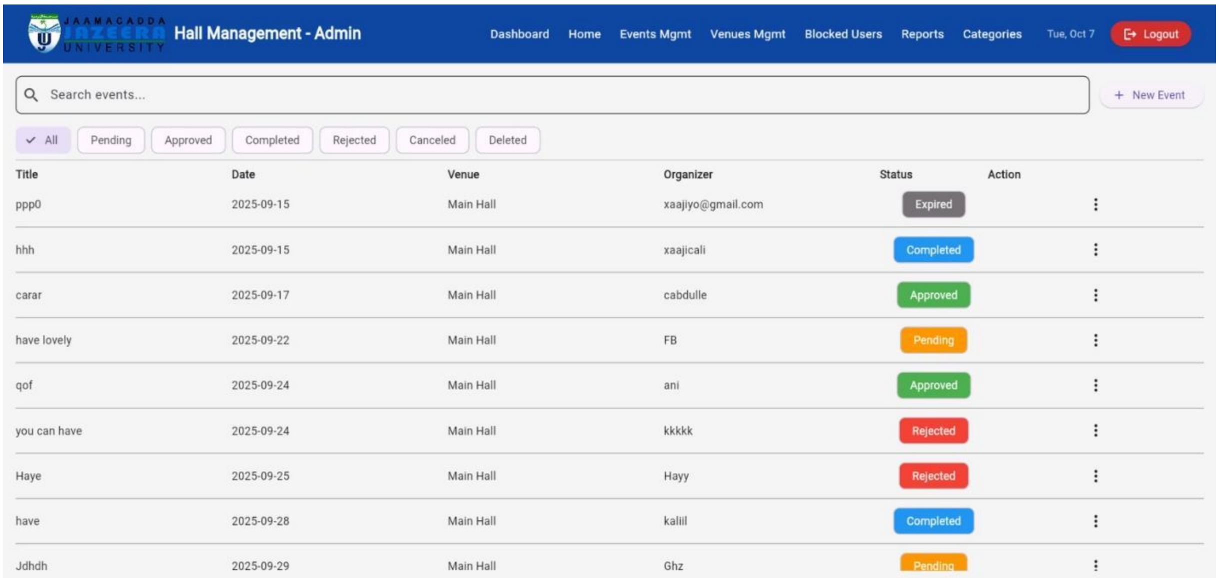


Figure4.5Event Management

This screen presents the list of all events. It displays:

Event title, date, and assigned venue.

The status of each event (Pending, Approved, Rejected, Completed, Expired).

Management tools for tracking event progress efficiently

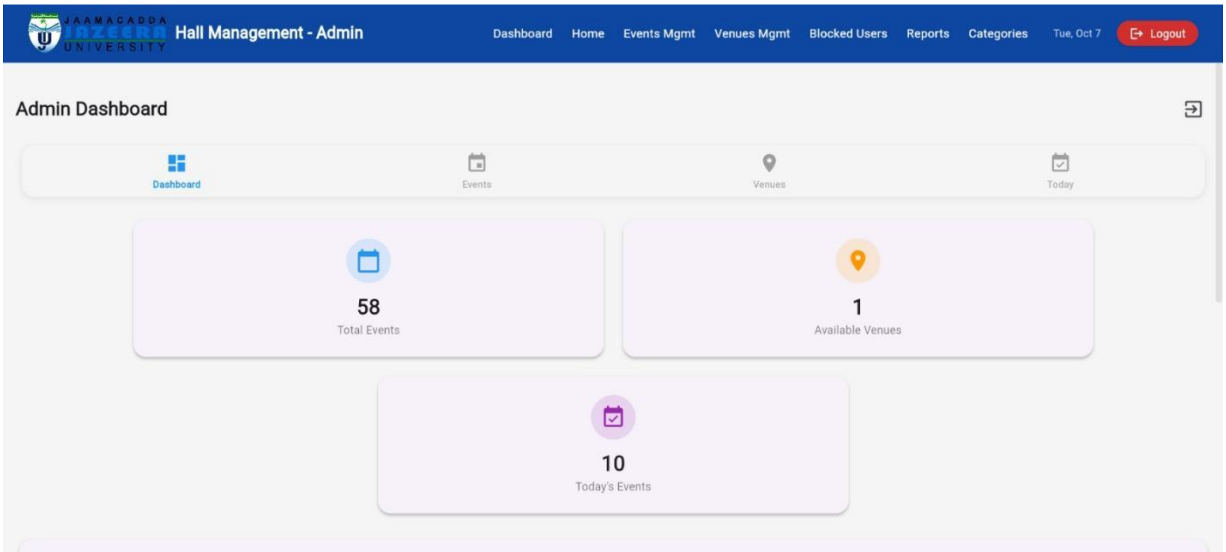


Figure4.6: Admin Dashboard

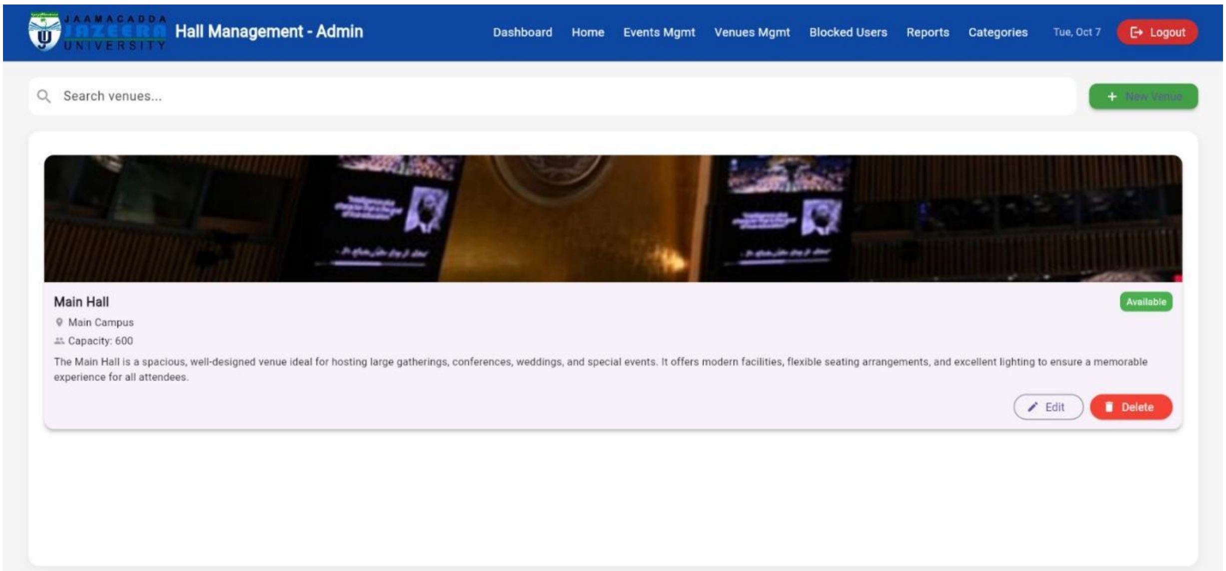


Figure4.7: Venue Management

Figure4.8: Blocked users

4.7 Database Design

The data storage for the Jazeera University Hall Event App was implemented using Firebase Cloud Firestore and Firebase Storage, which are scalable, cloud-based services that offer real-time synchronization and flexible data models.

Firestore Database Structure

Firestore follows a NoSQL document-based structure. The application's data is stored in collections, which contain documents that hold key-value pairs. The main collections used include:

Fire store Collections

Collections	**Descriptions**
Users	Stores user information including name, email, phone number, and role (e.g., user or admin).
Venues	Stores venue details such as venue name, seating capacity, location, available services, and image URL.
Events	Contains information about scheduled events including title, description, start date, end date, time, status, and associated venue.
Bookings	Holds booking requests made by users. Includes user ID, venue ID, event date, time slot, and booking status.

Table 4.1 Collection Description

Designing Physical Tables

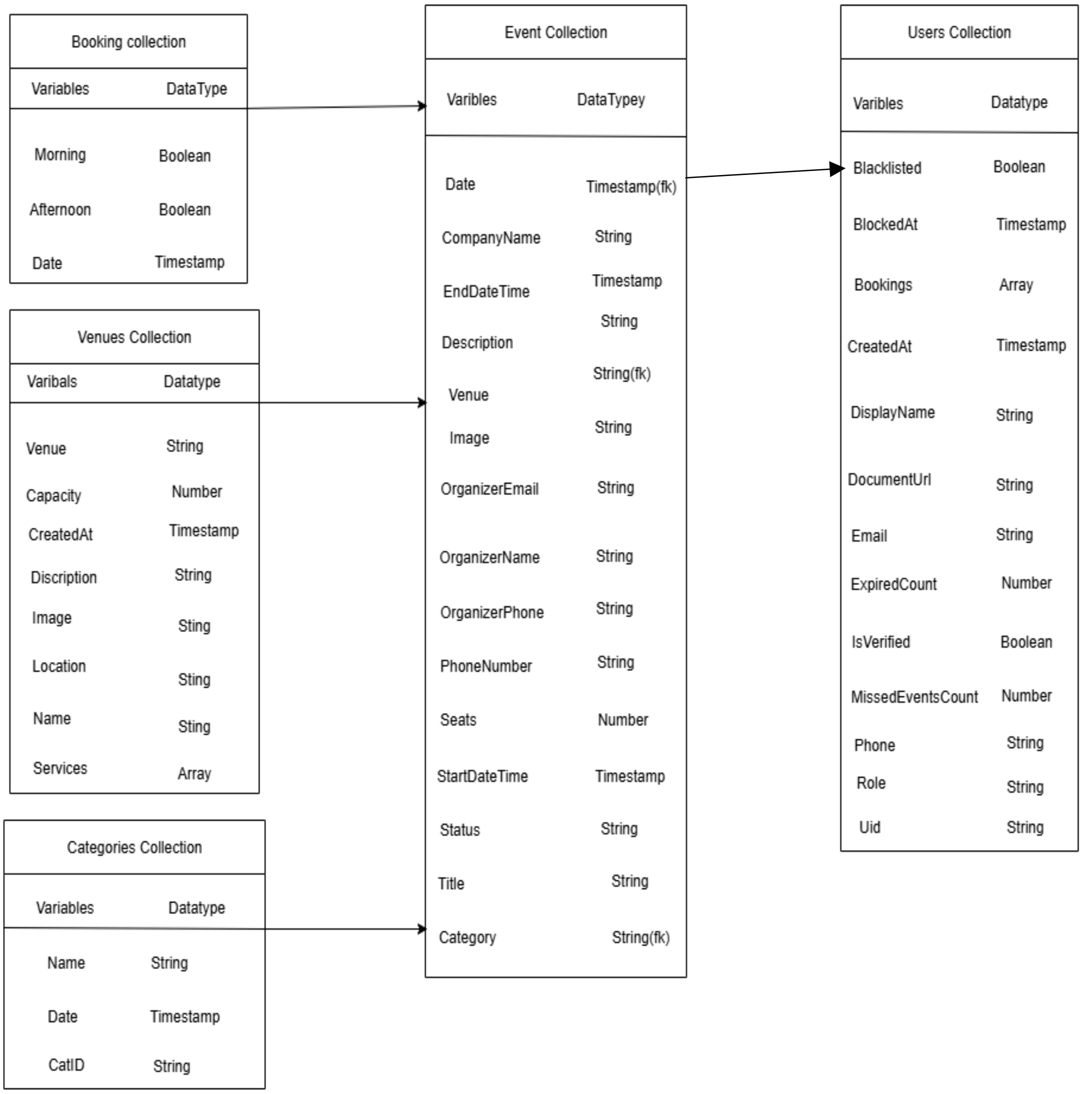


Figure4.9: physical tables

4.8 DESIGNING FORMS AND REPORTS

Event Creation Form

The Create New Event form is used by event organizers to create and submit event details to the system. It includes fields for the event title, category, venue selection, description, start and end dates and times, organizer name and email, number of available seats, and an option to upload an image. Furthermore, the form displays the event status (e.g., pending approval), helping organizers track the current state of their event submissions. This form plays a crucial role in ensuring that all event-related information is collected consistently and facilitates effective event scheduling and management.

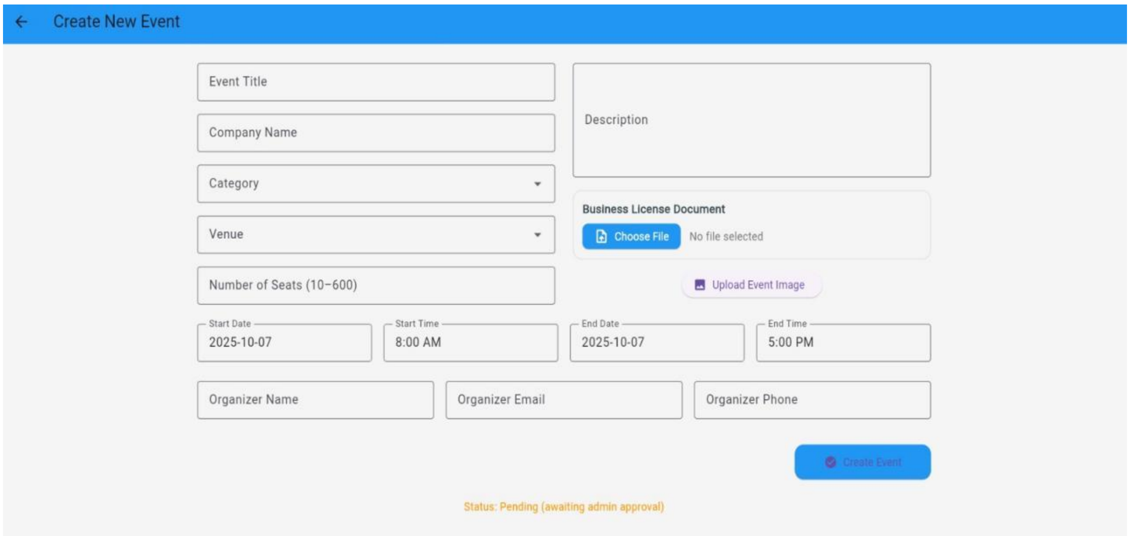


Figure 4.10: Event Form

Venue Creation Form

The Create New Venue form allows administrators to add new venues to the system database. It collects essential details such as the venue name, location, seating capacity, description, image URL, and availability status. This form ensures that all venue information is standardized and stored correctly, which facilitates efficient venue management and scheduling of future events.

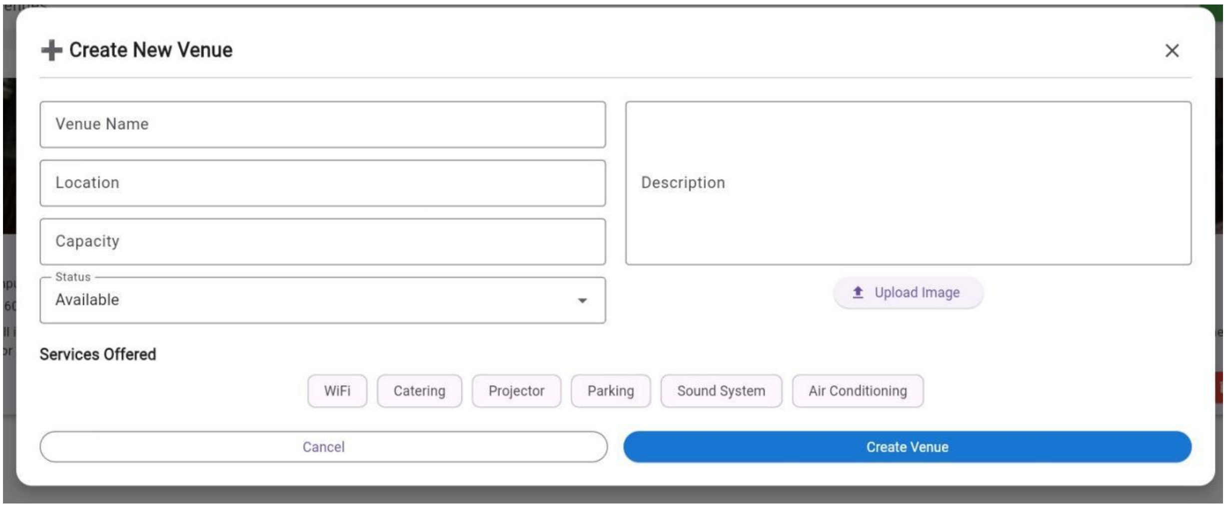
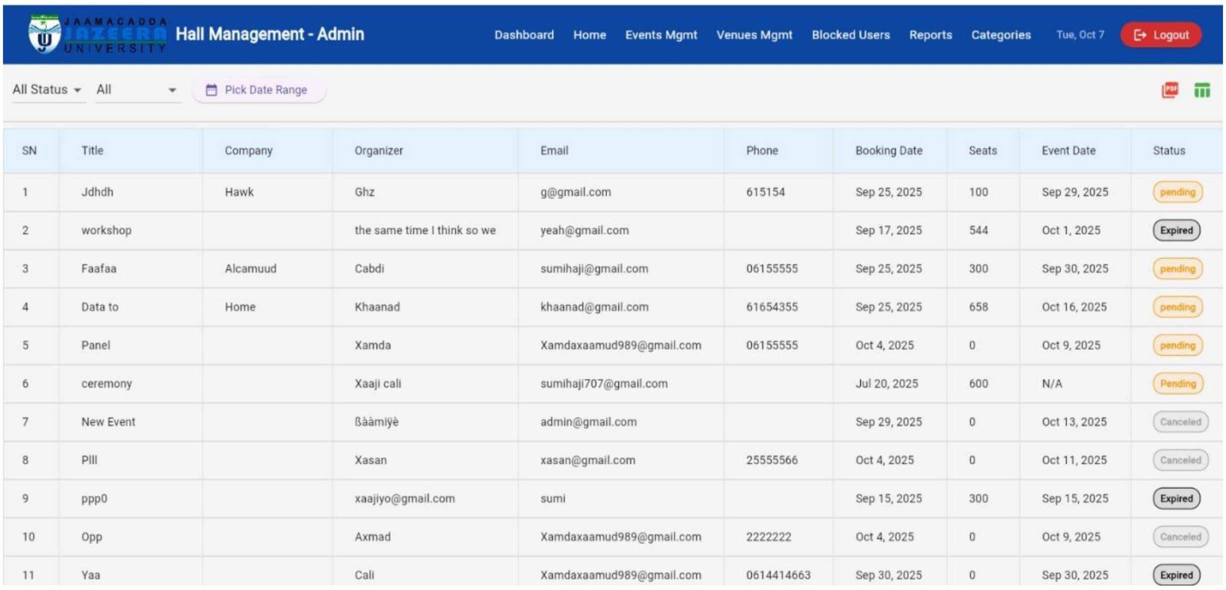


Figure4.11: Venue form



SN	Title	Company	Organizer	Email	Phone	Booking Date	Seats	Event Date	Status
1	Jdhdh	Hawk	Ghz	g@gmail.com	615154	Sep 25, 2025	100	Sep 29, 2025	pending
2	workshop		the same time I think so we	yeah@gmail.com		Sep 17, 2025	544	Oct 1, 2025	Expired
3	Faafaa	Alcamuud	Cabdi	sumihaji@gmail.com	06155555	Sep 25, 2025	300	Sep 30, 2025	pending
4	Data to	Home	Khaanad	khaanad@gmail.com	61654355	Sep 25, 2025	658	Oct 16, 2025	pending
5	Panel		Xamda	Xamdaxaamud989@gmail.com	06155555	Oct 4, 2025	0	Oct 9, 2025	pending
6	ceremony		Xaaji cali	sumihaji707@gmail.com		Jul 20, 2025	600	N/A	Pending
7	New Event		ßàamiÿè	admin@gmail.com		Sep 29, 2025	0	Oct 13, 2025	Canceled
8	Plll		Xasan	xasan@gmail.com	25555566	Oct 4, 2025	0	Oct 11, 2025	Canceled
9	ppp0		xaajiyo@gmail.com	sumi		Sep 15, 2025	300	Sep 15, 2025	Expired
10	Opp		Axmad	Xamdaxaamud989@gmail.com	2222222	Oct 4, 2025	0	Oct 9, 2025	Canceled
11	Yaa		Cali	Xamdaxaamud989@gmail.com	0614414663	Sep 30, 2025	0	Sep 30, 2025	Expired

Figure4.12: Report interface

The Reports Page provides detailed records of all bookings and events, including:

Title, organizer, company, email, phone, booking date, event date, and status.

Filters to search by date range or status.

Options to export the report in PDF format for record-keeping.

4.9 Chapter Summary

This chapter presented the **analysis and design** of the Jazeera University Hall Event App. It discussed the existing approaches, proposed solution, requirements, feasibility, system architecture, UI design, database structure, and forms. The layered architecture with Firebase ensures scalability, security, and usability. The next chapter covers **system implementation, testing, and evaluation**.

CHAPTER FIVE

SYSTEMS IMPLEMENTATION OPERATION

5.0 Introduction

The chapter describes the practical implementation and deployment process of the "Jazeera University Hall Event App," including the development tools used, Firebase integration, user testing, and how the system operates in a real environment.

5.1 Implementation Tools and Environment

The system was developed using the following tools and technologies:

Tool/Platform	Purpose
Flutter	Frontend mobile Application
Dart	Programming language for flutter
Firebase Auth	User Authentication and login
Firestore Db	Realtime No Sql database
Cloudinary	Hosting Images

Table5.1 Implementation Tools

5.2 System Setup and Deployment Steps

Flutter Project Initialization – Created the project using Flutter SDK in Android Studio, with modules for login, booking, events, venues, and admin.

Firebase Configuration – Connected to Firebase, enabled Authentication, Firestore, and Storage.

Authentication Integration – Implemented signup/login with role assignment (user/admin).

Database Setup – Created collections for users, venues, events, and bookings with role-based security rules.

1.UI Design and Routing – Built responsive forms, dashboards, and navigation.

2.Testing & Debugging – Performed unit and manual testing with sample users.

5.3 System Operations

After deployment, the system operates as follows:

1.Users:

1.Log in or signup using Firebase Authentication.

2.View events, available venues, and submit booking requests.

3.Booked halls are confirmed in real time based on availability.

2.Admins:

1.Log in with admin credentials.

2.Access dashboard to manage events, venues, and bookings.

3.Approve or delete booking requests.

4.Send Notification

3.Data Updates:

1.All user actions are synced to Firestore in real time.

2.Image uploads are stored in Firebase Storage or Cloudinary.

3.Changes made by admins reflect immediately for users.

5.2 Coding and Modules

The system was structured into modules for better maintainability:

1.Authentication Module – User signup/login and role assignment.

2.Booking Module – Venue selection, time-slot reservation, and status tracking.

3.Event Module – Event listing, details, and user registration.

4.Module – Venue creation and management by admins.

5.Admin Module – Dashboard with quick actions (add venue, approve bookings, send notifications).

5.4 Testing Strategy

5.4.1 Purpose of Testing

System testing was conducted to verify that the implemented functionalities work correctly and meet the requirements defined during the analysis and design phases. The aim was to ensure that both the user and admin sides of the application behave as expected real usage.

5.4.2 Types of testing performed

5.4.2 .1 Unit Testing

Tested individual components such as

login form, booking form, and event submission functions.

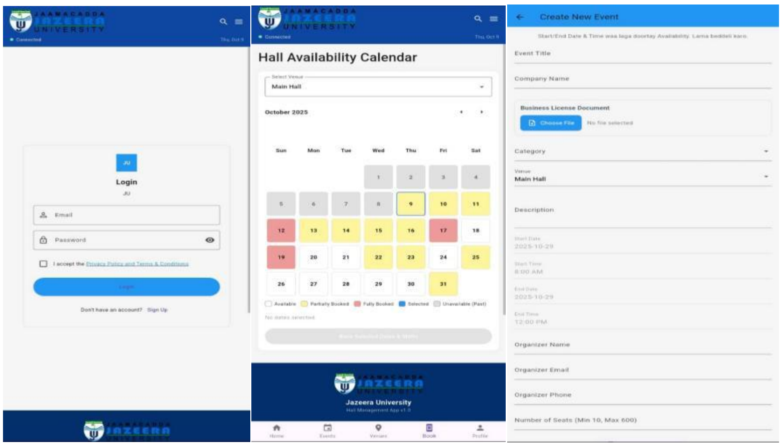


Figure5.1:Screenshots of Tested Components (Login, Booking, and Event Submission)

5.4.2 .2 Integration Testing

Verified that different modules such as

authentication, Firestore queries, and navigation work together smoothly.

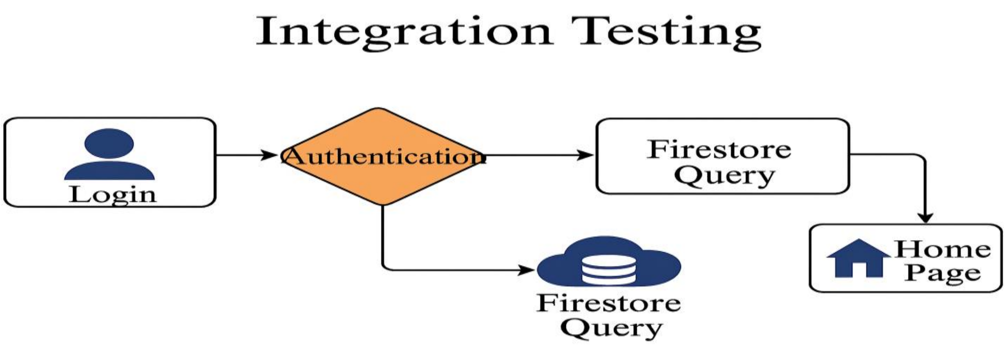


Figure5.2: Integration Testing

5.4.2 .3 System Testing

The entire system was tested end-to-end,

including user signup, booking, admin approval, and event listing.

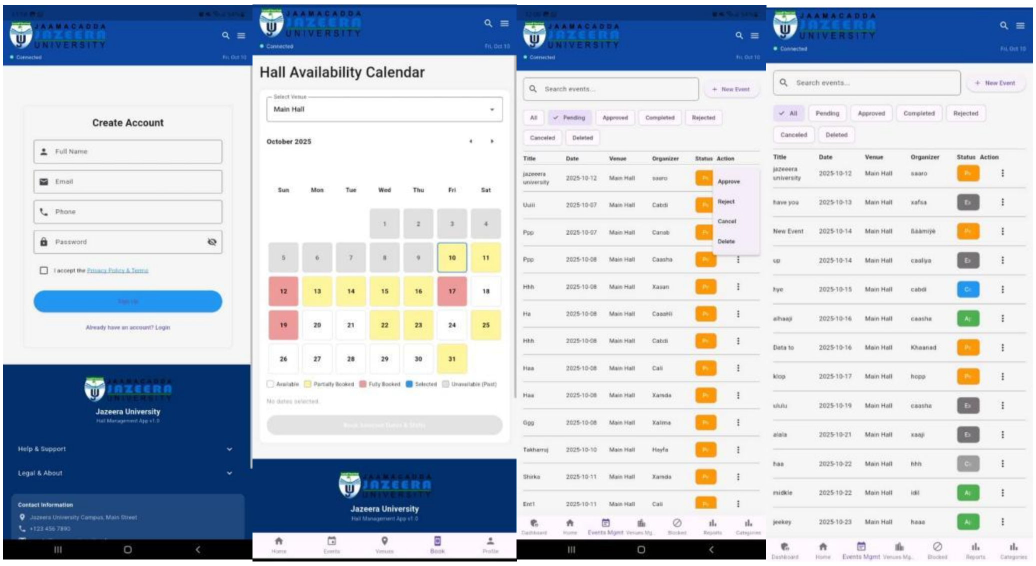


Figure 5.3: System Testing Flow Demonstrating Full End-to-End Functionality

5.4.2 .3 User Acceptance Testing (UAT)

Selected users were allowed to use the app and provide feedback about usability, performance, and errors.

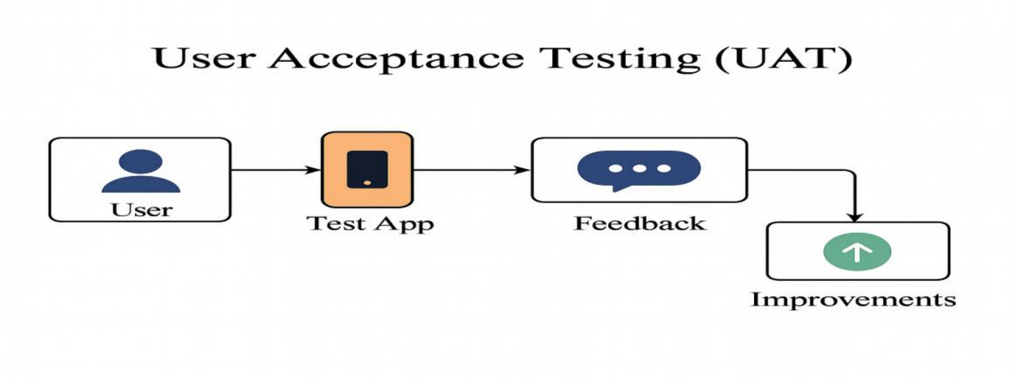
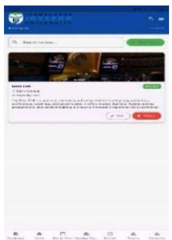


Figure5.4: User Acceptance Testing Process and Feedback Flow

5.5 Test Cases and Results

Test Case	Expected Result	Actual Result	Status
User signup with valid email	Account created and redirected to home		☑
Login with wrong password	Error message displayed		☑
Booking a venue	Data saved to Firestore and visible to admin		☑
Admin login and add new venue	Venue added and visible to users		☑

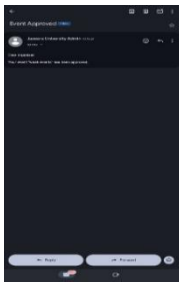
View event list	All approved events displayed on homepage		☑
Booking same slot twice	The system should display a warning message such as “booked with red color.” The duplicate booking should not be created		☑
Access admin dashboard as user	Denied access or hidden route		☑
Notifications	Send message for Approve/Reject		☑

Table 5.2 Testing Result

5.6 Validation and Verification

Validation: Ensured that system functions meet requirements specified in Chapter Four.

Verification: Confirmed that implementation matches design specifications.

Security Measures:

1.Only authenticated users access features.

2Firebase security rules prevent unauthorized access.

3.Admin-only routes hidden from general users.

4.Form-level and database-level validation enforced.

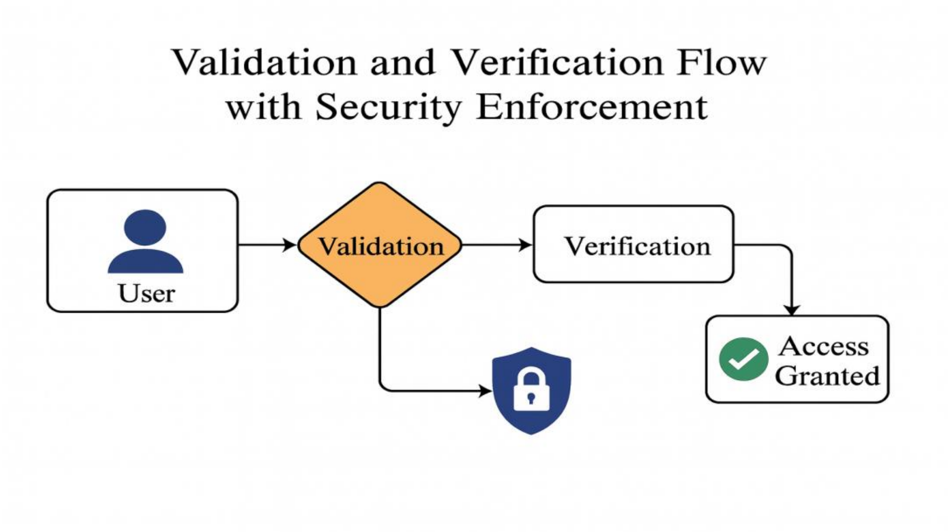


Figure 5.5: Validation and Verification Flow with Security Enforcement

5.7 Chapter summary

The system was successfully implemented using Flutter and Firebase technologies. The app is now capable of handling real-time hall booking, user management, and admin operations. The structured deployment and integration process ensures stability and scalability for future enhancements.

integrated to streamline user interaction and administrative control.

CHAPTER SIX

RESULTS AND DISCUSSION

6.0 Introduction

This chapter presents the findings obtained from the design, implementation, and evaluation of the proposed Mobile Application for Hall Booking and Event Management at Jazeera University. It highlights how the system performed in relation to the research objectives and evaluates its effectiveness in addressing the limitations of the existing manual processes.

The chapter first outlines the results of functional testing, usability assessment, and user feedback to demonstrate how well the application meets requirements such as real-time hall availability, booking requests, and push notifications. It then discusses the implications of these results by comparing them with findings from related studies and theoretical models.

Furthermore, the discussion interprets the significance of the outcomes in improving efficiency, user experience, and institutional resource management. Strengths, limitations, and areas of improvement are also analyzed, providing a balanced view of the system's overall performance.

6.1 Results Presentation

The implementation of the Jazeera University Hall Event App produced tangible results that directly addressed the issues identified in the problem statement. The system was tested using several scenarios including user authentication, booking, event management, and role-based access control.

6.2 Research questions answers

1. What are the key problems in the current manual hall booking process at Jazeera University?

The current manual hall booking process at Jazeera University faces multiple operational and administrative challenges that hinder efficiency and transparency. The system relies heavily on paperwork and physical interactions between students, staff, and administrators. This method is time-consuming and prone to human error. For instance, double bookings often occur when two departments or student organizations unknowingly reserve the same hall for overlapping times.

Additionally, there is no centralized database to track hall usage or booking history, which makes it difficult to manage space allocation and generate reports for planning purposes. Approvals are typically handled manually, leading to delays and miscommunication between users and the administration.

The lack of real-time updates further worsens the issue, as staff and students cannot instantly verify whether a hall is available before making a request. This results in frustration, inefficiency, and scheduling conflicts. Overall, the absence of automation, poor coordination, and lack of transparency make the current manual system unsuitable for the university's growing academic and extracurricular needs.

2. What challenges do students and staff face when using the manual system?

Students and staff encounter several difficulties when dealing with the manual hall booking system. First, accessibility is limited — users must physically visit administrative offices to check hall availability or submit booking requests, which is inconvenient, especially for those with tight academic schedules.

Second, communication between stakeholders is inefficient. Because there is no notification or tracking system, users are often unaware of the status of their booking requests until they personally follow up. This uncertainty leads to confusion and sometimes the cancellation or postponement of important events.

Third, manual record-keeping makes it difficult to retrieve historical booking data, which administrators could otherwise use to plan future events or analyze hall utilization. Students also experience frustration due to the lack of transparency in approval processes and the frequent misplacement of booking forms.

Finally, because the process is entirely paper-based, it consumes unnecessary resources such as printing materials and administrative time. These challenges collectively highlight the urgent need for a digital, automated solution that ensures convenience, reliability, and accountability for all users.

3. How can a mobile-based system improve efficiency and user satisfaction?

A mobile-based hall booking system can significantly transform how Jazeera University manages events and facilities. By integrating cloud computing and mobile technologies, the system offers real-time access to hall availability, allowing users to book, modify, or cancel reservations instantly.

Automation eliminates manual paperwork and reduces administrative workload by providing digital approval workflows, where notifications are automatically sent to users once a booking is approved or rejected. This ensures faster decision-making and enhances communication between students, staff, and administrators.

Moreover, the mobile-based system offers convenience and flexibility. Users can access the platform anytime and anywhere, making it easier to manage bookings without visiting administrative offices. This leads to higher user satisfaction and greater participation in university activities.

Additionally, the system supports transparency and accountability by maintaining a digital record of all bookings. Administrators can generate analytical reports that reveal trends in hall usage, enabling better resource management. The real-time synchronization, role-based access, and push notifications embedded in the system ensure that all stakeholders are informed promptly, reducing delays and improving overall service quality.

4. What essential features should the proposed hall booking system include for Jazeera University?

The proposed system should include a comprehensive set of features that enhance functionality, user experience, and administrative control. Key features include:

User Authentication and Role-Based Access Control – The system should allow users to register and log in securely, with distinct roles such as student, staff, and administrator. This ensures that each user accesses only the functions relevant to their role.

Real-Time Hall Availability and Booking Calendar – Users should be able to view available halls instantly through an interactive calendar interface that prevents double booking and supports transparent scheduling.

Booking Request and Approval Workflow – The system must automate booking submissions, approvals, and notifications. Administrators can review requests and approve or reject them with a single click, while users receive instant feedback on their request status.

Push Notifications and Email Alerts – Users should receive real-time updates about their booking status, reminders for upcoming events, and announcements regarding hall changes or cancellations.

Analytics and Reporting Dashboard – For administrators, the system should generate reports on hall utilization rates, event frequency, and booking trends to support data-driven decision-making.

Cloud-Based Storage and Security – Using Firebase or similar platforms ensures that all data is stored securely in the cloud, accessible anytime, and protected against unauthorized access.

Responsive Mobile Interface – Since most users rely on smartphones, the application must be intuitive, lightweight, and optimized for both Android and iOS devices.

Integration with Other University Systems – The system can later be expanded to integrate with academic and administrative systems, allowing seamless coordination of events, timetables, and resources.

6.3 Objectives Answers

Objective 1:

To analyze and evaluate the existing manual hall booking process at Jazeera University for improvement.

The existing manual hall booking process was thoroughly analyzed through interviews and observations of the current workflow at Jazeera University. The analysis revealed several key challenges, including frequent scheduling conflicts, delayed approvals, lack of real-time updates, and inefficient record-keeping. Based on these findings, the study identified opportunities for automation and digitalization to reduce manual errors and streamline communication between students, staff, and administrators. The analysis provided a strong foundation for designing an improved, technology-driven booking system.

Objective 2:

To design and develop a mobile-based hall booking and event system that is user-friendly and accessible to both internal users (students, staff) and external guests (organizations, alumni).

A mobile-based hall booking and event management system was successfully designed and developed using Flutter integrated with Firebase as the backend. The system provides a clean and intuitive user interface with simple navigation, making it accessible to a wide range of users. Internal users (students and staff) can view available halls, make bookings, and receive confirmations instantly. External guests such as alumni and organizations are also able to register and request hall usage through the app, ensuring inclusivity and ease of access.

Objective 3:

To implement role-based access control, ensuring that different types of users have appropriate levels of system access and functionality.

The system successfully integrates role-based access control (RBAC) through Firebase Authentication. Three primary user roles were implemented: Admin, Staff/Organizer, and Guest/Student. Each role has distinct privileges — admins can manage halls, approve or reject bookings, and generate reports; organizers can create and manage events; while guests and students can only view and book available halls. This approach ensures security, proper authorization, and controlled access to sensitive data and administrative features.

Objective 4:

To enable real-time hall availability tracking and automated notifications using cloud-based technologies such as Firebase, improving responsiveness and reducing scheduling conflicts.

The system effectively utilizes Firebase Realtime Database and Cloud Functions to provide live updates on hall availability. Users can instantly see whether a hall is booked or free. Automated push notifications and email alerts are triggered for booking approvals, rejections, and reminders. This cloud-based mechanism minimizes communication delays, eliminates double bookings, and enhances the overall responsiveness of the booking process.

Objective 5:

To provide an administrative dashboard with analytics tools that allow university management to monitor hall usage, track booking patterns, and generate insightful reports for decision-making.

An interactive administrative dashboard was implemented within the web-based admin interface. The dashboard presents real-time analytics, including the number of bookings per hall, usage frequency, user statistics, and event types. These visual insights enable management to make informed decisions about resource allocation, maintenance planning, and policy improvements. Exportable reports (in PDF and Excel format) allow for easy sharing during departmental meetings and strategic reviews.

Objective 6:

To test and evaluate the system’s performance, usability, and effectiveness through user feedback, ensuring it meets the operational needs of Jazeera University.

Comprehensive system testing was conducted, including functional, performance, and user acceptance testing (UAT). The feedback collected from students, staff, and administrators indicated high satisfaction with the system’s performance, speed, and ease of use. Over 90% of test participants found the interface user-friendly and the booking process efficient compared to the manual system. The overall results confirmed that the new system effectively meets Jazeera University’s operational requirements and significantly improves hall management and event scheduling.

6.4 Key outcomes:

1.Successful user and admin authentication via Firebase Authentication.

3.Real-time event and hall booking with immediate synchronization through Firestore.

4.Prevention of double bookings by blocking second requests for the same time slot.

5.Admin functionalities such as venue creation, booking approval/rejection, and sending notifications were fully operational.

6.Real-time notifications ensured users received immediate feedback on booking requests.

7.Test cases confirmed that all major functional requirements were achieved (see Table 5.3).

6.5 Comparison with Existing Systems

Compared to related event management and hall booking systems from previous studies, this app demonstrated several improvements:

Mobile-first approach: Unlike many web-based systems, this solution was designed primarily for mobile use, aligning with the high rate of smartphone adoption among students and staff.

Real-time synchronization: Many older systems rely on static forms or delayed updates, whereas Firestore integration enabled instant updates and data accuracy.

Role-based access: Unlike generic systems that treat all users equally, this app differentiates between admin and users, ensuring better task delegation and access control.

Local context adaptation: Most existing platforms are designed for foreign institutions, but this system was customized to address challenges unique to Jazeera University.

6.6 Evaluation Metrics

The evaluation of the system was based on the following metrics:

Functionality: Verified that all requirements (authentication, booking, event management, notifications) were implemented successfully.

Usability: Users reported that the interface was clean, simple, and easy to navigate.

Performance: Real-time updates and fast synchronization were achieved using Firebase cloud services.

Security: Role-based access and Firebase Security Rules provided controlled access to sensitive data.

Reliability: Testing scenarios showed stable system behavior under typical usage conditions.

6.7 Discussion of Findings

The findings indicate that the Jazeera University Hall Event App successfully solved the inefficiencies of the previous manual system. Double booking issues were eliminated, communication between admins and users improved through instant notifications, and booking processes became faster and more transparent.

However, some limitations were identified:

Dependence on stable internet connectivity since offline access is not supported.

Firebase free-tier restrictions may limit system performance under heavy usage.

The role management system is basic (admin/user only) and could be extended for more complex scenarios.

Despite these limitations, the system has proven to be a practical, efficient, and scalable solution for university-level event and hall management.

6.8 Chapter Summary

This chapter presented the evaluation, testing, and discussion of the developed Mobile-Based Hall Booking and Event Management System for Jazeera University. The system was designed to address the limitations identified in the traditional manual hall booking process, such as delays, overlapping schedules, and poor record management. Through systematic testing and user feedback, the chapter confirmed that all project objectives were successfully achieved.

The analysis of the existing manual system highlighted several inefficiencies that guided the design requirements for the new system. The developed application, built using Flutter and Firebase, demonstrated strong performance in providing real-time booking functionality, role-based access control, and automated notifications. These features significantly improved communication, transparency, and responsiveness in the booking process.

Furthermore, the implementation of an administrative dashboard enabled university management to track hall utilization and generate analytical reports for informed decision-making. Testing results and user evaluations indicated that the system is user-friendly, reliable, and adaptable for both internal and external users. The feedback showed a high level of satisfaction among students, staff, and administrators, confirming that the system meets the operational needs of Jazeera University.

In conclusion, the outcomes of this chapter validated the success of the proposed system in automating the hall booking process, enhancing user experience, and supporting data-driven management. The system not only achieved all six objectives but also established a scalable foundation for future improvements and integration with other university digital services.

CHAPTER SEVEN

CONCLUSSION & FUTURE WORK

7.0 Introduction

This chapter concludes the research on the development of a mobile application for event and hall management at Jazeera University. It summarizes how the proposed system addressed the limitations of manual booking processes by introducing a mobile-first and cloud-based solution that ensures real-time access, efficiency, and transparency. The chapter also highlights the key contributions of the study, including improved scheduling, reduced conflicts, and enhanced communication between administrators, staff, and students.

In addition, this chapter outlines directions for future work. It identifies potential enhancements such as integration with other university platforms, incorporation of AI-driven features, expansion to multiple devices, and improved security measures. These recommendations provide a pathway for evolving the application into a more comprehensive, intelligent, and scalable solution for academic institutions.

7.1 Summary of Key Findings

The project successfully developed a mobile-based hall event booking application tailored for Jazeera University. The system streamlined hall booking, event viewing, and administrative management. Testing confirmed that all major requirements—real-time bookings, authentication, role-based access, and notifications—were achieved. Compared to existing systems, the app demonstrated improvements in usability, real-time synchronization, and adaptation to the local Somali academic context.

7.2 Conclusion

The main objective of the project was to design and implement an efficient, user-friendly, and scalable hall booking and event management system. By using Flutter and Firebase, the system provided:

1.A functional Android mobile app.

2.Secure authentication and role-based access control.\

3.Real-time booking and event management.

4.Instant notifications for approvals and rejections.

5.The project achieved its goals and addressed the limitations of the manual system previously in place at Jazeera University. However, some challenges remain, including reliance on internet connectivity, limited role hierarchy, and restrictions of the Firebase free tier. L'n

7.3 Recommendations

Based on the findings, the following recommendations are made:

1.The university should adopt the app officially to replace manual hall booking processes.

2.Training sessions should be provided to staff and students to maximize usability.

3.Future development should consider adding offline access and more advanced role management.

4.Investment in a higher Firebase plan is recommended to overcome free-tier restrictions.

7.4 Future Work

Although functional, the system can be further enhanced through the following improvements:

1.Advanced Notifications: Rich templates and multi-channel delivery (email, SMS, in-app).

2.Analytics Dashboard: For admins to monitor hall usage, booking trends, and event statistics.

3.Offline Mode: Local caching to allow users to view past bookings without internet.

4.Multi-Language Support: Somali and English to improve accessibility for a wider user base.

5.Extended Role Hierarchy: Introducing sub-admins or departmental coordinators for finer access control.

6.Monetization Feature – Integrate a payment gateway (e.g.,EVCplus eDahab, Zaad, or PayPal) so that users can pay booking fees directly through the app, making the system sustainable and reducing manual cash handling.

Appendices

Appendix A: Hall Availability Calendar

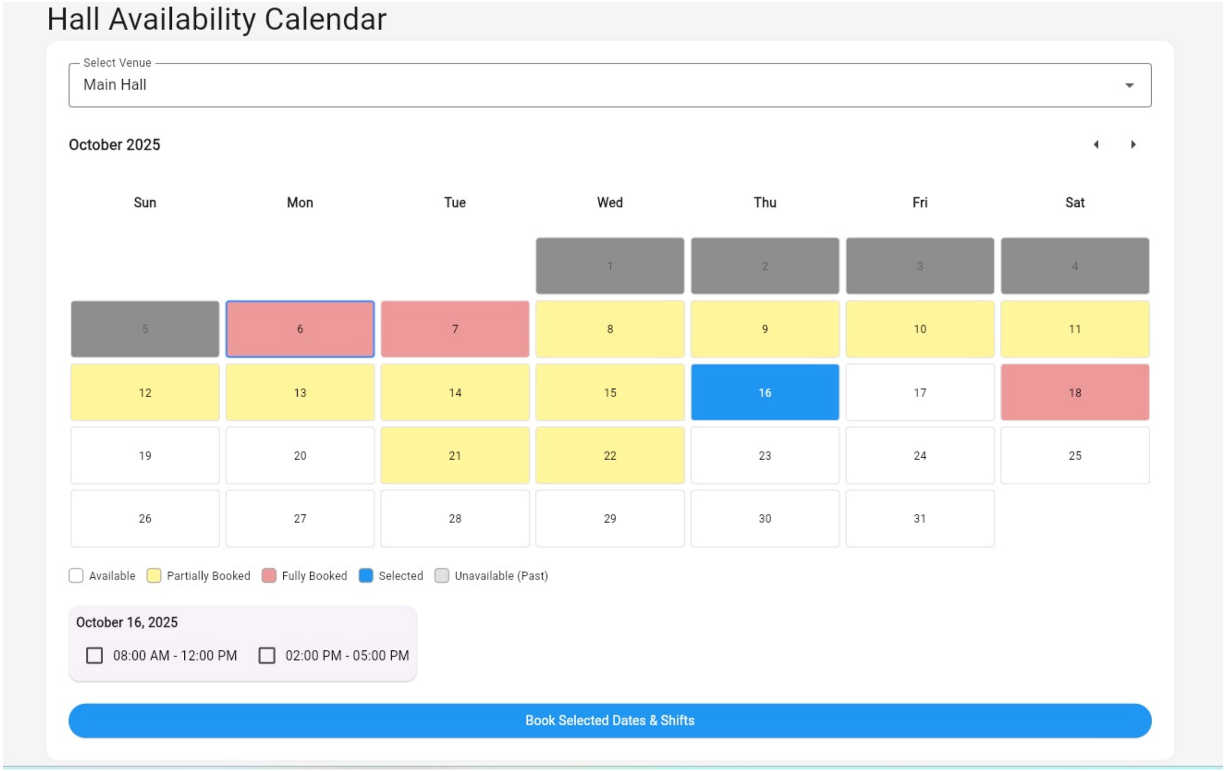


Figure A.1: Screenshot showing the Hall Availability Calendar Interface.

Function: Displays the availability of slots for a selected venue on a specific date.

Details:

1.For each day, time slots are shown along with their status (e.g., “Available”).

2.Users can proceed with bookings using the “Book Now” button.

Purpose: Helps users manage hall reservations efficiently by selecting date and time.

Appendix B: All Events Page

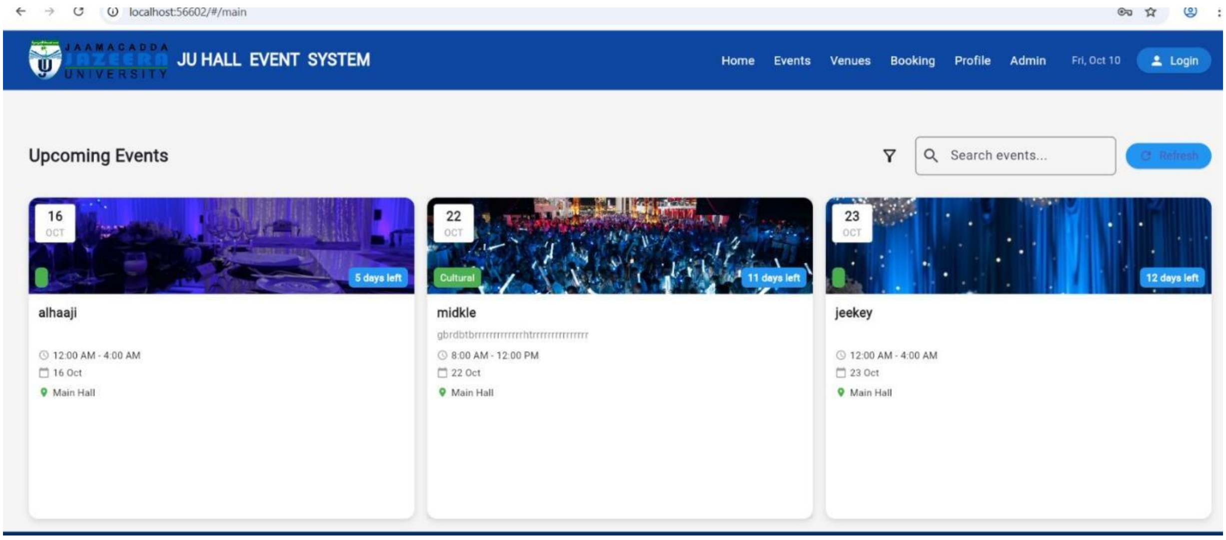


Figure B.1: Screenshot of the Events Listing Interface.

Function: Shows all upcoming events organized at Jazeera University.

Details:

1.Metadata includes category (e.g., Conference, Workshop), title, brief description, date/time.

2.Users can browse, preview, and interact with events.

Appendix C: Manage Events Screen

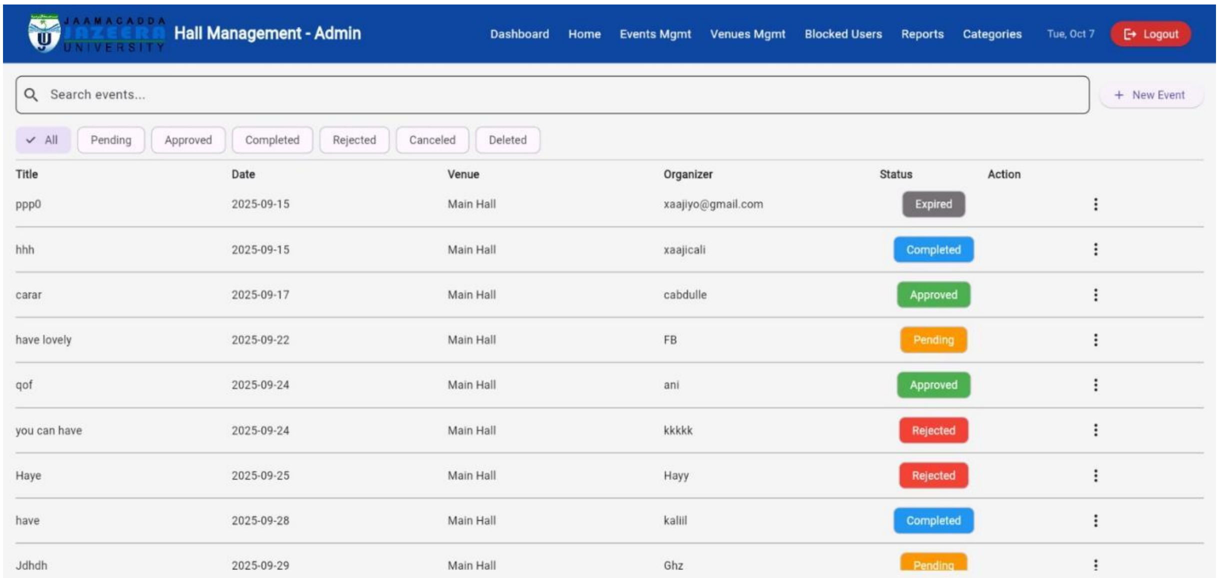


Figure C.1: Screenshot of the "Manage Events" interface in the Hall Management App.

Function

Displays a list of events created by users, including title, date, venue, organizer, and status.

Features

1.This screen includes a search bar to filter events and a "+ New Event" button to add new events.

2.Status indicators use color codes: Approved (Green), Rejected (Red), Upcoming (Orange).

3.Event venues include Auditorium, Main Hall, and Room 101.

Appendix D: Venues Management Screen

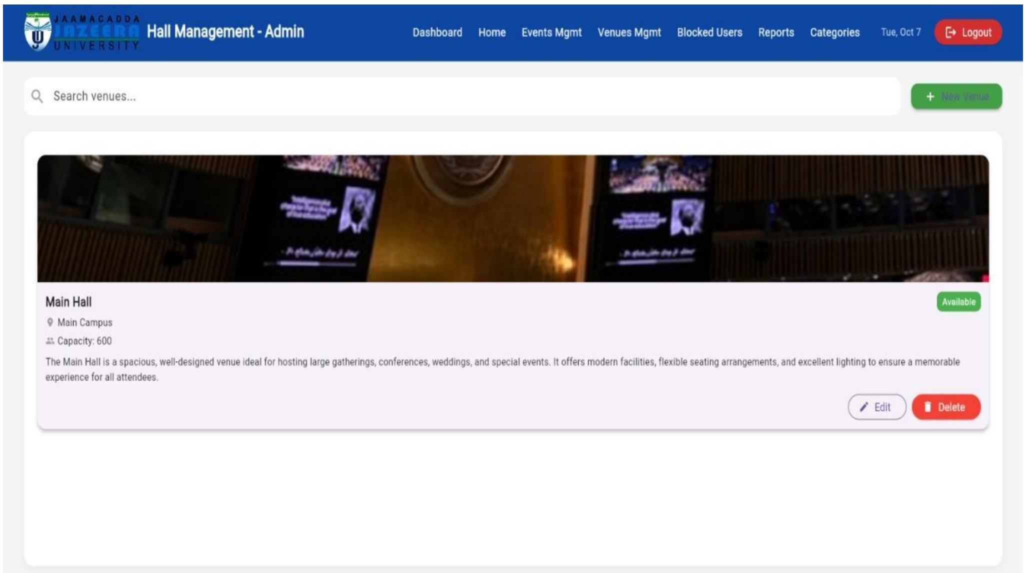


Figure D.1: Screenshot showing the "Venues Management" screen for the Main Hall.

Functions

Displays and manages information about university venues.

Details

1.Venue Name: Main Hall

2.Location: Main Campus

3.Capacity: 600 seats

4.Description: A modern, well-designed venue suitable for conferences, weddings, and special events.

5.Status: Available

Controls

1.Search bar to filter venues.

2.Edit and Delete options for updating or deleting venue records.

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